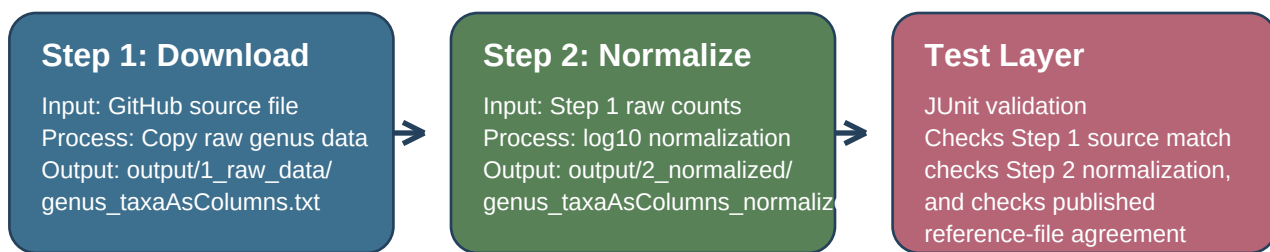


OrchestraSimpleDemo Test Report

Generated on April 12, 2026

This report summarizes the current automated validation status for the OrchestraSimpleDemo bioinformatics pipeline.

Current Orchestration Pattern



The pipeline currently orchestrates a raw-data acquisition step followed by normalization, with JUnit tests validating source fidelity, normalized-value correctness, and agreement with the published reference spreadsheet.

Test Summary

This report summarizes the Java/JUnit testing completed so far for the OrchestraSimpleDemo bioinformatics pipeline. The current automated checks cover raw-data integrity, normalized-output correctness, and agreement with the published normalized reference spreadsheet.

Test ID	Scope	Result	Evidence
T1	Step 1 raw-data validation	PASS	The file output/1_raw_data/genus_taxaAsColumns.txt exactly matched the
T2	Step 2 normalization validation	PASS	Every numeric cell in output/2_normalized/genus_taxaAsColumns_normaliz
T3	Step 2 reference-file validation	PASS	The local normalized table matched the published genus_taxaAsColumnsL

Execution Notes

The JUnit suite was executed successfully on April 12, 2026. One test downloads the canonical source file from GitHub at runtime and compares the remote bytes with the local Step 1 output bytes. A second test recomputes normalized values from the raw count table and verifies every numeric cell in the Step 2 output. A third test compares the local normalized spreadsheet with the published normalized-with-metadata reference file after skipping metadata columns.

Reference source:

https://raw.githubusercontent.com/kwinglee/UrbanRuralChina/master/16SrRNA/inputData/RDP/genus_taxaAsColumns.txt

Reference normalized file: https://raw.githubusercontent.com/kwinglee/UrbanRuralChina/master/16SrRNA/inputData/RDP/genus_taxaAsColumnsLogNorm_WithMetadata.txt

Current Coverage

Coverage now includes both current orchestration steps: verification that the downloaded raw data file is an exact mirror of the expected source, and verification that normalized output values satisfy the stated formula across the entire table, plus confirmation that the local normalized spreadsheet agrees with the published reference file on shared taxa columns. Future tests can extend this to end-to-end pipeline execution and additional downstream stages.